

ONLINE ACTIVITY #2 — PERSONALIZED LINUX & GIT CHEAT SHEET

1. BASIC LINUX COMMANDS

Command	Sample usage
pwd	pwd → show current folder path
ls	ls -la → list files, including hidden files
cd	cd Documents → enter a folder
mkdir	mkdir project → create a folder
touch	touch index.html → create an empty file
cp	cp a.txt backup.txt → copy a file
mv	mv old.txt new.txt → move/rename a file
rm	rm notes.txt → remove a file
cat	cat notes.txt → display file contents
grep	grep "error" log.txt → find matching text
head / tail	head -n 5 file.txt / tail -n 5 file.txt → first/last 5 lines
chmod	chmod +x script.sh → add execute permission

2. BASIC GIT COMMANDS

Command	Sample	My description	Common options / flags
git init	git init	Start Git in the current folder.	-q, -b
git restore	git restore file.txt	Bring a file back from the index/another commit; useful for undoing local edits.	--staged, -s, -p
git status	git status -sb	Check changed, staged, and untracked files.	-s, -b
git branch	git branch feature	List branches or create a new one.	-a, -d, -m
git add	git add index.html	Put chosen changes into the staging area.	., -A, -p
git checkout / git switch	git switch main git checkout main	Change branches. <i>switch</i> is the clearer modern choice for branch switching.	-c (switch), -b (checkout)
git commit	git commit -m "Update page"	Save the staged snapshot with a message.	-m, -a
git merge	git merge feature	Combine another branch into the current branch.	--no-ff, --abort
git log	git log --oneline	View the commit history.	--oneline, --graph, -p
git tag	git tag v1.0	Mark a specific point in project history.	-a, -d, -l
git diff	git diff	Compare changes that are not staged yet.	--staged, --stat, -U
git clone	git clone	Copy an existing remote repository to your computer.	--depth, -b

3. GIT'S THREE-TREE ARCHITECTURE

WORKING DIRECTORY	INDEX / STAGING AREA	HEAD / REPOSITORY
Where I edit files. It is my working copy and sandbox.	The changes I selected for the next commit. Updated by git add .	My last committed snapshot/history. HEAD points to the current branch tip.

Simple flow: edit file → **git add** → staged change → **git commit** → new snapshot. Think: **Working Directory** → **Index** → **HEAD/history**.

References: Git Documentation — Reference and command manuals (git-scm.com/docs); Git Book, “Reset Demystified — The Three Trees” (git-scm.com/book/en/v2/Git-Tools-Reset-Demystified); GNU Coreutils Manual (gnu.org/software/coreutils/manual). Accessed September 21, 2026.